

Sequential Task:

Performance Analysis of Operations on Java Collections

Object-Oriented Programming and GUI

April 16, 2026

1. Copy the provided *.java files into your own project.
2. Then run the `Main.java` file. The program will display measured execution times (in milliseconds) for the `Problem1Test` class and the `ArrayList` collection.
3. In the `Problem1Test` file, locate the `execute` method and analyze its behavior.
4. Based on the analysis, formulate a one-sentence description of the operation performed by this method.
5. Next, open the attached spreadsheet and verify whether your proposed description is consistent with the information contained in the file.
6. Enter the obtained results into the spreadsheet.
7. Repeat the above steps for other collections for which it is possible to perform the operations defined in the `execute` method.
8. In each row of the spreadsheet, highlight the shortest execution time (the optimal collection) using a green color.
9. Perform the procedure for all `Problem*` files, and then attach the completed spreadsheet as the result of the task.